

Woojun Jung

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RESEARCH INTERESTS

Representation Learning: learning representations from heterogeneous data and interaction histories for adaptive memory and long-horizon personalization.

Human-AI Interaction: designing conversational AI that helps people articulate their needs, understand information, and make informed decisions.

EDUCATION

Korea University

M.S. in Computer Science and Engineering

Advisor: Prof. [Susik Yoon](#)

Mar 2025 – Feb 2027 (Expected)

Korea University

B.E. in Computer Science and Engineering

Mar 2022 – Feb 2025

Korea University

B.F.A. in Industrial and Information Design

2 years of Military Service: Republic of Korea Air Force (2017 – 2019)

Mar 2016 – Feb 2022

Relevant Coursework: CS/AI - Algorithms (A+), Computer Architecture (A), Artificial Intelligence (A), Deep Learning (A), Optimization Theory (A), Reinforcement Learning (A), Natural Language Processing (A), Brain and Machine Cognition (A+); Design/HCI - Interaction Design (A+), Interface Design (A+), Design and Human Factors (A+), Design Methodology (A+).

PUBLICATIONS

**: equal contribution*

- [3] **Segment-driven Structural Induction and Semantic Alignment for Heterogeneous Tabular Representation**
[Woojun Jung](#) and [Susik Yoon](#)
ICML 2026 — [\[arXiv\]](#) [\[Code\]](#)
- [2] **Can Structural Cues Save LLMs? Evaluating Language Models in Massive Document Streams**
[Yukyung Lee](#), [Yebin Lim*](#), [Woojun Jung*](#), [Wonjun Choi](#), [Susik Yoon](#)
KDD 2026 — [\[ACM DL\]](#) [\[Code\]](#)
- [1] **HAETAE: In-Domain Table Pretraining with Header Anchoring**
[Woojun Jung](#) and [Susik Yoon](#)
SIGIR 2025 (Short) — [\[ACM DL\]](#) [\[Code\]](#)

RESEARCH EXPERIENCE

DAIS Lab, Korea University

Research Intern (Aug 2024 – Feb 2025); Research Assistant (Mar 2025 – Present)

Aug 2024 – Present

Calibrated User Beliefs for Conversational Agents

May 2026 – Present

- Formulating long-horizon personalization as maintaining calibrated beliefs over context-dependent and evolving user preferences rather than reducing them to fixed textual point memories.
- Designing pilot comparisons across deterministic text memories and probabilistic preference states.
- Exploring whether representation-derived uncertainty can be calibrated against subsequent user feedback and used to guide when agents should apply a preference or ask for clarification.

Structured Understanding in Evolving Information Streams

KDD 2026

- Developed the Temporal QA task, one of three benchmark tasks alongside Topic Clustering and Summarization.
- Designed and implemented its evaluation protocol, benchmark dataset, and human-annotation guidelines.

Structured Understanding of Tabular Data

ICML 2026; SIGIR 2025

- Led two first-author projects on domain-specific tabular representation learning.
- Proposed header anchoring to regularize contextualized header embeddings toward context-independent anchors, stabilizing semantics across row contexts and schema variants.
- Developed a pretraining method that jointly learns schema and column contexts with entropy-routed objectives to preserve domain-coherent and instance-specific attribute semantics.

APPLIED RESEARCH COLLABORATION

Foundation Models for Enterprise Relational Data, KT (Korea Telecom)

Mar 2025 – Present

- Collaborated with KT on foundation-model methods for enterprise relational databases.
- Analyzed challenges in adapting tabular ML to organizational data with heterogeneous schemas.
- Prototyped a pretraining pipeline for schema-aware attribute representations for enterprise tables.
- Investigating schema-linking methods that map natural-language questions to relevant tables and columns.

SELECTED PROJECTS

Dr.Snap: Multimodal Dermatology Decision-Support System

2024

- Designed and built a conversational prototype that combines skin photos and symptom descriptions to provide preliminary guidance before a clinic visit.
- Developed a dialogue workflow that uses retrieved condition information and prior user responses to generate follow-up questions and narrow candidate conditions.

Bianca: Full-Scale Interactive Refrigerator Prototype

2020

- Developed Arduino code and electronics and built a full-scale refrigerator prototype that maps handle-press locations to corresponding illuminated regions on the door.

TEACHING EXPERIENCE

Teaching Assistant, Data Science, Korea University

Spring 2025

- Prepared and delivered tutorials on LaTeX and visualization with matplotlib for undergraduate students.

Teaching Assistant, Artificial Intelligence, Korea University

Fall 2025

- Prepared and delivered a tutorial on multimodal representation learning with CLIP for undergraduate students.

AWARDS & SCHOLARSHIPS

Master's Fellowship (KRW 12,000,000), National Research Foundation of Korea (NRF)

2025–2026

Second Place, Korea University AI Tech Day Poster Presentation

2025

Third Place, Korea University Capstone Design Competition

2024

Third Place, Samsung Design Competition

2019

SKILLS

Programming: Python, C, SQL, Git, GitHub, LaTeX, HTML, CSS, JavaScript

ML & Systems: PyTorch, CUDA, Transformers, vLLM, NumPy, pandas, scikit-learn, matplotlib

Design & Hardware Prototyping: Figma, Illustrator, Photoshop, Rhinoceros, KeyShot, 3D printing, Arduino

Languages: Korean (Native), English (Fluent)